INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II, Fourth Semester Examination, March - 2025

Subject Code: BCACC402

Subject Name: Programming with Java

Full Marks: 30 Set No: 1 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1. What is the use of super in java? 2
2. Can we access subclass members if we create an object of superclass? 2
3. Why a top level java class cannot be private? 2
4. Discuss static variable with proper example. 2
5. Explain 'this' keywords with suitable example. 2
6. Apply the throws keyword in a method declaration and explain its significance with an example. 2

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. Write down the difference between final and finally in Java. 5
8. "Interface provide full abstraction." Justify. 5
9. Explain how BufferedReader improves efficiency compared to Scanner for reading input. Provide an exampic program to demonstrate this. 5

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Explain what exceptions are in Java. Differentiate between checked and unchecked exceptions with examples. Write a simple Java program that uses a try-catch block to handle Arithmetic Exception. 10
11. Explain the different stages of the thread lifecycle in Java. Write a JAVA program to demonstrate the concept of Main thread. 10

--End--

INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II, Fourth Semester Examination, March - 2025

Subject Code: BCACC402

Subject Name: Programming with Java

Full Marks: 30 | Set No: 2 | Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. What are the Exception Handling Keywords in Java? 2
2. Can we inherit private members of base class to its subclass? 2
3. How do Applets differ from Application programming? 2
4. How can method overriding be implemented in Java? Give an example. 2
5. What is Abstract class? 2
6. Demonstrate the use of try catch in Java with a simple example that handles division by zero. 2

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Can we overload methods on run time? Create an application in Java to implement dynamic method dispatch. 5
8. Write the declaration syntax of packages. Write a Java program to illustrate user defined packages. 5
9. Analyze the advantages and disadvantages of multithreading over multitasking in Java. Provide a real-world example. 5

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Write a Java program to create a class Vehicle with attributes like brand and speed. Create two subclasses: Car and Electric Car with additional attributes.

    Implement a method to display the details of each vehicle, showcasing multilevel inheritance. 10

11. Evaluate the input and output streams in Java's file handling mechanism. How do these streams enhance file manipulation operations? 10

--End--

INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II, Fourth Semester Examination, March - 2025

Subject Code: BCACC402

Subject Name: Programming with Java

Full Marks: 30 Set No: 3 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1. "public static void main(String[ ] args)", Why we write public static? 2
2. Write few names of built in packages in JAVA. 2
3. Is it possible to have methods with definition in interface? 2
4. Explain how a String object can be created by string literals. 2
5. Can we make the abstract methods static in Java? 2
6. Apply the throws keyword in a method declaration and explain its significance with an example. 2

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. Discuss the importance of interface in JAVA. Explain with appropriate examples. 5
8. Explain how naming collision can be avoided by packages. 5
9. Explain how BufferedReader improves efficiency compared to Scanner for reading input. Provide an example program to demonstrate this. 5

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Explain the life cycle of a Java applet with a diagram. Describe how the init(), start(), stop(), and destroy() methods work. 10
11. Develop a JAVA based application to demonstrate how to create and manipulate child thread using Runnable interface. 10

--End--